Chicago Taxi Fare Analysis

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Group Name:
The Last Semicolon

I. Intro

Chicago's Department of Business Affairs and Consumer Protection provided monthly reports of all taxi trips in 2019. The dataset is filtered on the first quarter of 2019 with 3343 observations. It consists of the following attributes: **date**; **day type** (weekday or weekend); **fare**; **tip**; **payment type** (cash or credit); travel **distance** in miles; travel **duration** in seconds; **speed** in mph.

We will look at whether taxi fare amounts are predictable given the distance traveled, duration spent, and speed traveled. This supports making a future prediction of the impact of any predicted variable change on total city cab fares. This analysis helps understand how operational factors such as trip distance and duration influence taxi pricing, which is useful for both city regulators and riders.

We will also look at tipping. Since tips are not recorded if a customer uses cash, we will only look at the customers with a recorded tip. We would like to test if the mean common US average tip etiquette of 15% is the same or different.

II. Multiple Linear Regression

To understand what drives the taxi fare amount, we started by fitting a multiple linear regression model. We aimed to understand how the fare relates to the distance, duration, and speed as predictors. Figure 1 shows the model equation as $\widehat{fare} = 7.62 + 3.03(distance) + 0.00082(duration) - 0.22(speed)$. For a one-mile increase in distance, the estimated fare increases by $3.036, holding duration and speed constant. For a one mph increase in speed, the estimated fare amount decreases by $0.22, holding distance and duration constant.

We also included the Variance Inflation Factor (VIF) test to determine if we have some multicollinearity between our predictors since speed is derived from distance and duration (speed = distance/duration). Looking at the result from the model, our overall

model fit R^2 adjusted is 0.8634. About 86.34% of sample variation in the taxi fare amount can be explained by a linear relationship of the taxi fare amount as a function of distance, duration, and speed.

The overall model hypothesis test has a null hypothesis that distance, duration, and speed have no linear relationship with fare. The alternative hypothesis is that at least one of these variables is associated with fare. The model's F-value is 7043.82 and p-value is less than 0.0001. We reject the null hypothesis and conclude that there is statistical evidence that at least one of these variables is associated with fare.

Then looking at the variables independently, the null hypothesis would be that the beta variable equals zero and the alternative is that the beta variable does not equal zero. All the variables have a p-value less than 0.0001. Thus, there is statistical evidence that each individual variable is significant.

While statistical significance can sometimes be affected by multicollinearity, the VIF for all predictor variables is below 5, suggesting that multicollinearity is not a serious concern in this model. Therefore, the coefficient estimates are considered reasonably stable and reliable for interpretation.

Figure 2 shows that the assumptions are not met. Linearity assumption is not met since the points are not randomly scattered. Constant variance assumption is not met since the points are not evenly spread out and there may be heteroscedasticity. Normality is not met since the points on the Q-Q plot deviated from the diagonal, especially in both tails. Lastly, our data shows some highly influential outliers. One point had a leverage of 0.8. This model violates key assumptions and may not be appropriate for inference. Looking more into the variables may help us fit a better model.

III. Multiple Linear Regression with Transformed Variables

We created histograms for predictor variables (fare, distance, duration, speed) to see the distribution of the data. Figure 3 shows the distribution of fare, distance and duration are heavily right-skewed, with the normal curve fitting poorly. Speed is less strongly right skewed compared to the other variables, but still shows noticeable positive skew, with only an approximate normal fit. To address the skewness, we applied log transformation on both dependent and independent variables to see if we can improve the model. Figure 4 shows the regression results. Since all the variables are right-skewed and exhibit fan-shaped residual patterns, this suggests that the scale of impact when changing a variable is a multiplicative relationship with fare rather than an additive one. The R^2 adjusted is 82.69% which is lower than the first multiple linear regression model. This decrease is expected, as improving model assumptions can come at the cost of slightly reduced explanatory power. The assumptions are still not met, but it did improve the constant variance assumption. Normality also improved. They still exhibit a mild fan shape, which is an improvement over the previous model. However, it introduced severe multicollinearity with VIFs exceeding 25,000 among the independent variables. This is extreme and destabilizing. This could affect the reliability of our regression coefficients since distance, trip duration, and speed are closely connected. For example, if a trip covers a long distance, it will typically also take more time, and speed is directly determined by both. This means that longer trips and time spent in traffic, such as construction season, are key drivers of higher fares. Because of this, the model struggles to determine whether fare increases are due to distance, time, or speed individually.

Therefore, we decided to use ridge regression to help minimize the multicollinearity effect. We started by fitting ridge with all the log-transformed variables and gave ridge k values from 0 to 0.005, increasing by 0.001, see Figure 5. Proc print outputted the ridge regression parameter estimates table showing lambda, RMSE, and parameter estimates. Lambda = 0.002 is the best one to use because it provides a balance between reducing multicollinearity and maintaining a low RMSE (0.31022). The final model is $(\widehat{lnfare}) =$

$-0.6542 + 0.3(lnDistance) + 0.49(lnDuration) - 0.06(lnSpeed)$. This is a better model than the multiple regression model that we tried earlier.

Looking at the VIF for $ln(fare)$, at $\lambda = 0.002$ the VIF of the distance dropped to 3.09, where VIF < 5. The standardized coefficient confirms that estimates were driven by multicollinearity rather than real relationships. From our table output, $\lambda = 0.002$ had lower RMSE among values where VIF fell below 5. Therefore, it was selected for our ridge k-value giving us a ridge model that we can use to predict the fare amount.

Then we changed the ridge value to 0.002 and included outseb in the proc reg line to get the standard error for each parameter estimate, see Figure 6. The standard errors are small relative to the magnitude of the coefficients, suggesting that the estimates are reasonably stable after applying ridge regression. The RMSE is 0.30594 which is good because it indicates that the model's predicted values are closer to the actual, observed data points. The final equation is

$$\widehat{(lnfare)} = -0.6542 + 0.29999(lnDistance) + 0.48715(lnDuration) - 0.06053(lnSpeed)$$

The final model suggests that a 1% increase in distance is associated with approximately a 0.30% increase in fare, holding other variables constant. Similarly, a 1% increase in trip duration increases fare by about 0.49%, while speed has a small negative relationship with fare. The negative relationship with speed may reflect that higher speeds are associated with less congested trips, resulting in shorter durations and therefore lower fares. In future analysis, we will look at the model without including speed as a predictor variable. We include it now for completeness.

IV. Predictability Verification

To validate predictive performance, we will check the prediction table in Figure 7. This will show how close the prediction value is to the actual one and if all observations fall within their respective 95% prediction intervals. For example, taking observation number

8, first we need to exponentiate the model's log fare back to the original dollar scale. Actual fare amount = $e^{3.04}$ = \$20.90; predicted fare amount = $e^{2.9792}$ = \$19.67; which is very close to the actual value. This falls within 95% CI predicted (2.3791, 3.5793) = (\$10.79, \$35.84). This demonstrates that the model provides reasonably accurate predictions on the original dollar scale.

V. Student's T-Test Analysis

We wanted to see when customers tip, do they tip the common average tip etiquette of 15% or is it different. We previously graphed the distribution of tip percentage, see Figure 8Figure 8. The average tip percentage is 15.98% and standard deviation is 4.24%. To validate this tip percentage, we had a null hypothesis that the mean tip percentage is equal to 15%. The alternative hypothesis was the mean tip percentage was different than 15%. Using the Proc Ttest method where H0=0.15 and filtering on only customers with a recorded tip, we have a t-value of 9.3 and p-value < 0.0001, see Figure 9. So, we reject the null hypothesis (H_0) and conclude there is strong statistical evidence that the mean tip percentage for customers who tip is different from 15%.

The assumptions for the two-sided t-test are met. The normal curve of the distribution of tip percentage fits the data well. The highest bar shows about 30% of observations fall within that range around 15%. The Q-Q plot shows almost all the points fitting the line. The assumptions are met, and in the future, we would like to do a one-tailed, upper test to see if the mean tip percentage is greater than 15%. As a practical tip consideration, cash tips are not recorded or available, which suggests that cash tips may be at a higher percentage.

Then we looked at when customers tip, do they tip the same on the weekday vs the weekend or is it different. Using the Proc Ttest method with class day type and filtering on only customers that tip, the equality of variance has a F value of 1.03 and p-value of 0.6618, see Figure 10. Since the p-value is greater than alpha=0.05, the difference in

variances is not significant and will use pooled variance. The pooled variance has a t-value of 0.4 and p-value of 0.6918. We fail to reject H0 and conclude there is no statistical evidence that the mean tip percentage for customers who tip is different between weekdays and weekends. The assumptions are met. The histograms show the data fitting the normal curve. The Q-Q plots show the data fitting the line.

VI. Conclusion

Duration and distance are the primary drivers for Chicago cab fare amounts. Speed was considered, but its individual effect is difficult to interpret due to multicollinearity. Ridge was the best model at lambda = 0.002, since the predicted values are closer to the actual, observed data points. In the future, we will remove the predictor variable speed to see how distance and duration predict fare amount. From a practical standpoint, cab fare is cost driven: primarily driven by trip distance and duration, which are the key factors influencing cost which makes the impact of rate changes predictable.

Mean tip percentage is statistically significantly different from the common average tip etiquette of 15%. In the future, we will look to see if the mean tip percentage is greater than 15%. Also, there is no statistical evidence of a difference to the tip percentage between weekday and weekend fares.

These findings may be useful for the City of Chicago in understanding factors that influence taxi fares and evaluating fare structures. From a driver's perspective, since no significant difference in tipping behavior was found between weekdays and weekends, tip percentages appear to be relatively consistent across day types. However, tipping behavior is likely influenced by multiple factors beyond fare alone, so increasing fares alone does not necessarily guarantee higher tips.

VII. Appendix

I. Outputs

Figure 1

The REG Procedure
Model: MODEL1
Dependent Variable: fare

Number of Observations Read	3343
Number of Observations Used	3343

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	835513	278504	7043.82	<.0001
Error	3339	132020	39.53881		
Corrected Total	3342	967533			

Root MSE	6.28799	R-Square	0.8635
Dependent Mean	17.11902	Adj R-Sq	0.8634
Coeff Var	36.73101		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	1	7.62073	0.22789	33.44	<.0001	0
distance	1	3.03568	0.03741	81.15	<.0001	3.70644
duration	1	0.00081857	0.00010491	7.80	<.0001	1.38931
speed	1	-0.22833	0.01997	-11.43	<.0001	3.05852

Figure 2

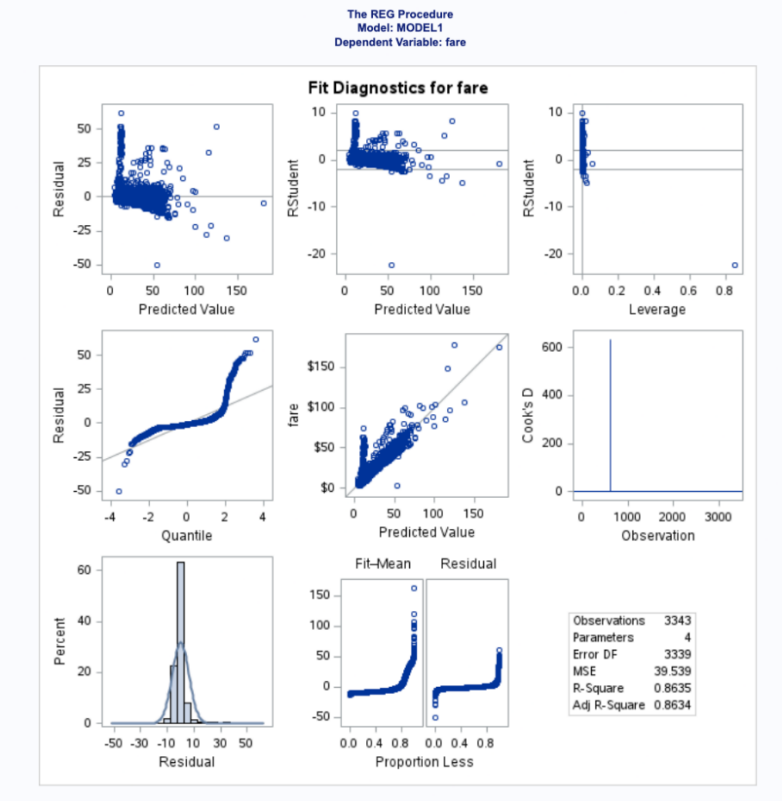
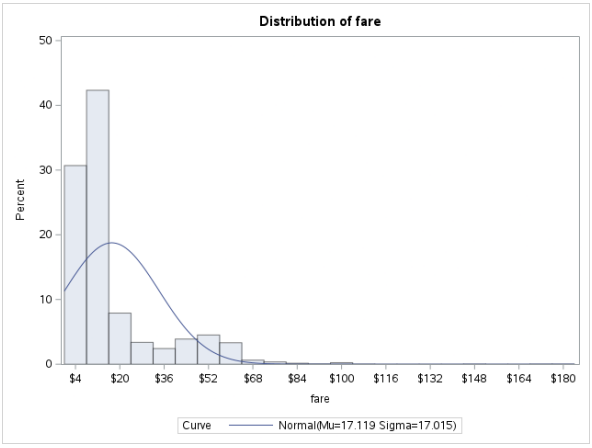
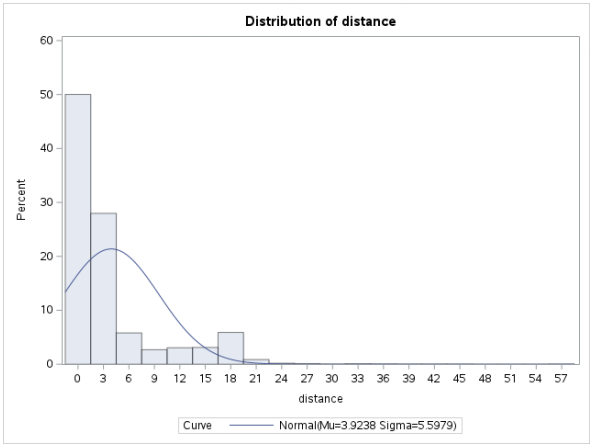
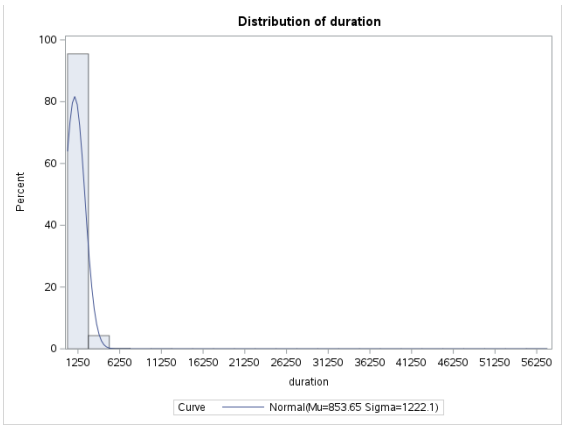

The REG Procedure
Model: MODEL1
Dependent Variable: fare
Fit Diagnostics for fare
Residual
Predicted Value
RStudent
Predicted Value
RStudent
Leverage
Residual
Quantile
fare
Predicted Value
Cook's D
Observation
Percent
Residual
Fit–Mean
Residual
Proportion Less
Observations 3343
Parameters 4
Error DF 3339
MSE 39.539
R-Square 0.8635
Adj R-Square 0.8634

Figure 3


Distribution of fare
Percent
fare
Curve
Normal(Mu=17.119 Sigma=17.015)


Distribution of distance
Percent
distance
Curve
Normal(Mu=3.9238 Sigma=5.5979)



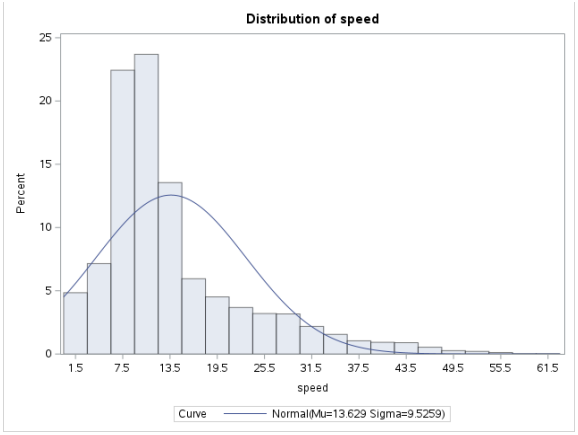


Figure 4

The REG Procedure
Model: MODEL1
Dependent Variable: lnFare

Number of Observations Read	3343
Number of Observations Used	3343

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	1492.57580	497.52527	5315.42	<.0001
Error	3339	312.53163	0.09360		
Corrected Total	3342	1805.10743			

Root MSE	0.30594	R-Square	0.8269
Dependent Mean	2.51742	Adj R-Sq	0.8267
Coeff Var	12.15300		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	1	54.50173	5.69127	9.58	<.0001	0
lnDistance	1	7.04007	0.69538	10.12	<.0001	25280
lnDuration	1	-6.24247	0.69444	-8.99	<.0001	9999.80085
lnSpeed	1	-6.81318	0.69668	-9.78	<.0001	10989

The REG Procedure
Model: MODEL1
Dependent Variable: lnFare

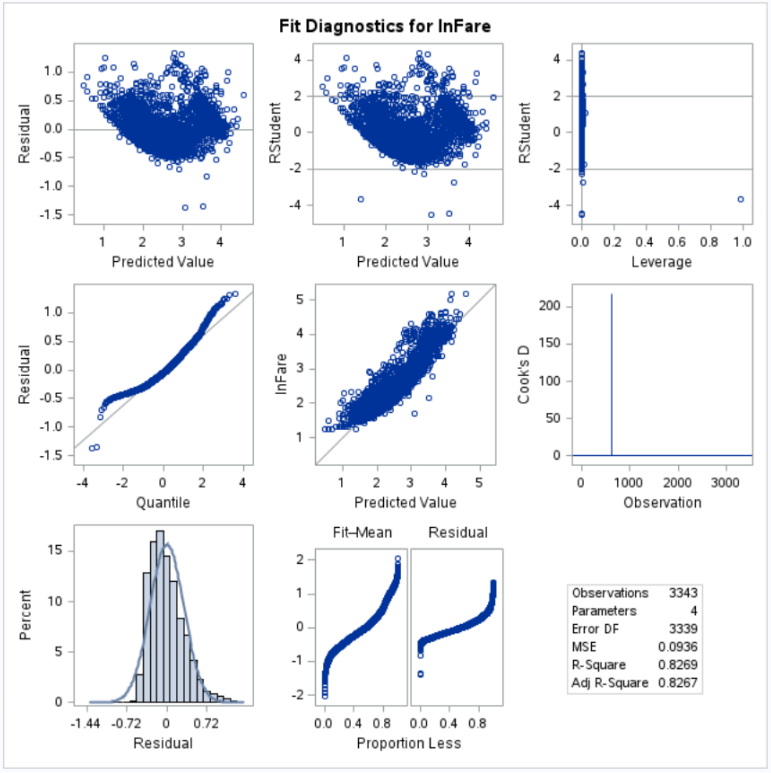


Figure 5

The REG Procedure
Model: MODEL1
Dependent Variable: lnFare

Number of Observations Read	3343
Number of Observations Used	3343

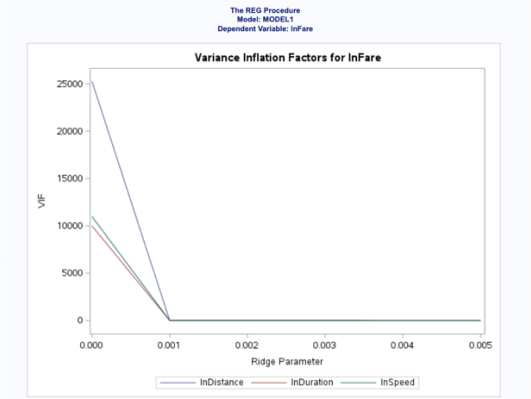
Analysis of Variance

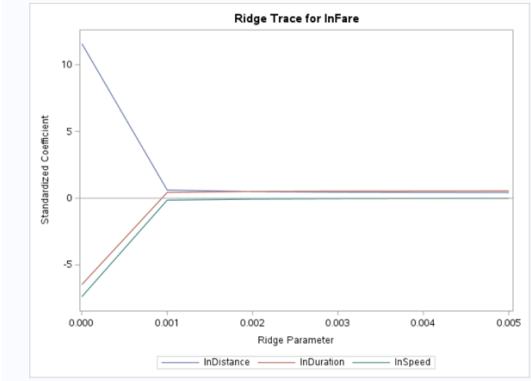
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	1492.57580	497.52527	5315.42	<.0001
Error	3339	312.53163	0.09360		
Corrected Total	3342	1805.10743			

Root MSE	0.30594	R-Square	0.8269
Dependent Mean	2.51742	Adj R-Sq	0.8267
Coeff Var	12.15300		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	1	54.50173	5.69127	9.58	<.0001	0
lnDistance	1	7.04007	0.69538	10.12	<.0001	25280
lnDuration	1	-6.24247	0.69444	-8.99	<.0001	9999.80085
lnSpeed	1	-6.81318	0.69668	-9.78	<.0001	10989





Obs	_MODEL_	_TYPE_	_DEPVAR_	_RIDGE_	_PCOMIT_	_RMSE_	Intercept	lnDistance	lnDuration	lnSpeed	lnFare
1	MODEL1	PARMS	lnFare	.	.	0.30594	54.5017	7.04	-6.24	-6.81	-1
2	MODEL1	RIDGEVIF	lnFare	.000	.	.	.	25280.34	9999.80	10989.11	-1
3	MODEL1	RIDGE	lnFare	.000	.	0.30594	54.5017	7.04	-6.24	-6.81	-1
4	MODEL1	RIDGEVIF	lnFare	.001	.	.	.	11.52	5.24	5.65	-1
5	MODEL1	RIDGE	lnFare	.001	.	0.31013	-0.0733	0.37	0.42	-0.13	-1
6	MODEL1	RIDGEVIF	lnFare	.002	.	.	.	3.09	1.91	1.99	-1
7	MODEL1	RIDGE	lnFare	.002	.	0.31022	-0.6542	0.30	0.49	-0.06	-1
8	MODEL1	RIDGEVIF	lnFare	.003	.	.	.	1.50	1.28	1.29	-1
9	MODEL1	RIDGE	lnFare	.003	.	0.31025	-0.8488	0.28	0.51	-0.04	-1
10	MODEL1	RIDGEVIF	lnFare	.004	.	.	.	0.94	1.05	1.05	-1
11	MODEL1	RIDGE	lnFare	.004	.	0.31027	-0.9453	0.26	0.52	-0.02	-1
12	MODEL1	RIDGEVIF	lnFare	.005	.	.	.	0.67	0.95	0.93	-1
13	MODEL1	RIDGE	lnFare	.005	.	0.31028	-1.0023	0.26	0.53	-0.02	-1

Figure 6

The REG Procedure
Model: MODEL1
Dependent Variable: lnFare

Number of Observations Read	3343
Number of Observations Used	3343

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	1492.57580	497.52527	5315.42	<.0001
Error	3339	312.53163	0.09360		
Corrected Total	3342	1805.10743			

Root MSE	0.30594	R-Square	0.8269
Dependent Mean	2.51742	Adj R-Sq	0.8267
Coeff Var	12.15300		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	54.50173	5.69127	9.58	<.0001
lnDistance	1	7.04007	0.69538	10.12	<.0001
lnDuration	1	-6.24247	0.69444	-8.99	<.0001
lnSpeed	1	-6.81318	0.69668	-9.78	<.0001

Obs	_MODEL_	_TYPE_	_DEPVAR_	_RIDGE_	_PCOMIT_	_RMSE_	Intercept	lnDistance	lnDuration	lnSpeed	lnFare
1	MODEL1	PARMS	lnFare	.	.	0.30594	54.5017	7.04007	-6.24247	-6.81318	-1
2	MODEL1	SEB	lnFare	.	.	0.30594	5.6913	0.69538	0.69444	0.69668	-1
3	MODEL1	RIDGE	lnFare	.002	.	0.31022	-0.6542	0.29999	0.48715	-0.06053	-1
4	MODEL1	RIDGESEB	lnFare	.002	.	0.31022	0.0702	0.00780	0.00973	0.00950	-1

Figure 7

The REG Procedure
Model: MODEL1
Dependent Variable: lnFare

Output Statistics

Obs	Dependent Variable	Predicted Value	Std Error Mean Predict	95% CL Predict		Residual
1	2.30	2.1766	0.007481	1.5766	2.7767	0.1260
2	3.90	3.3088	0.0117	2.7085	3.9091	0.5932
3	3.57	2.9981	0.0199	2.3970	3.5992	0.5714
4	2.88	1.8098	0.0138	1.2094	2.4103	1.0722
5	1.70	1.8635	0.007821	1.2635	2.4636	-0.1588
6	1.50	1.3766	0.0126	0.7762	1.9769	0.1275
7	2.20	1.8147	0.009048	1.2146	2.4148	0.3825
8	3.04	2.9792	0.009168	2.3791	3.5793	0.0653
9	1.56	1.3621	0.0146	0.7615	1.9626	0.1961
10	2.14	2.3376	0.007041	1.7376	2.9376	-0.1975

Figure 8

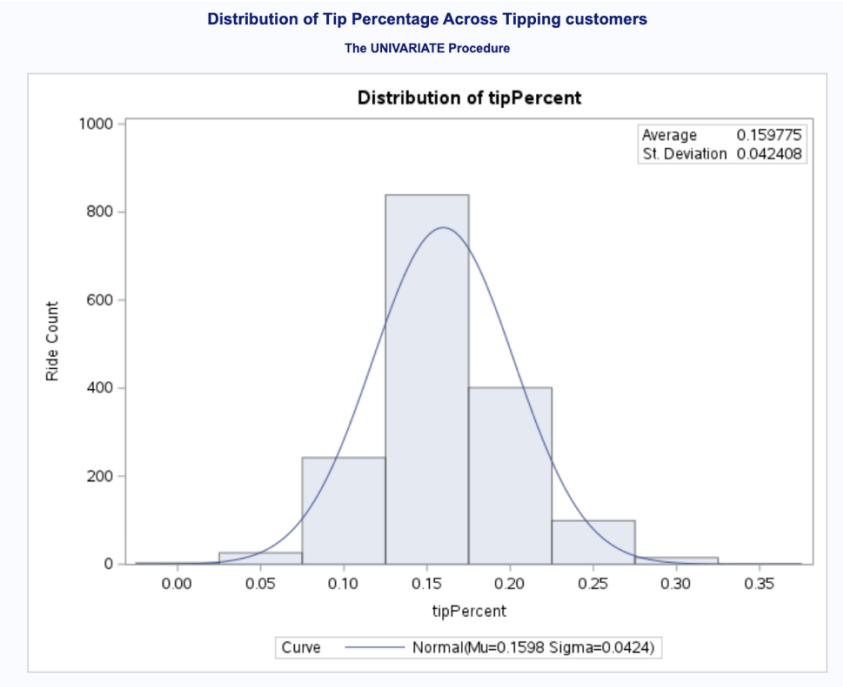

Distribution of Tip Percentage Across Tipping customers
The UNIVARIATE Procedure
Distribution of tipPercent
Average 0.159775
St. Deviation 0.042408
Ride Count
1000
800
600
400
200
0
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
tipPercent
Curve
Normal(Mu=0.1598 Sigma=0.0424)

Figure 9

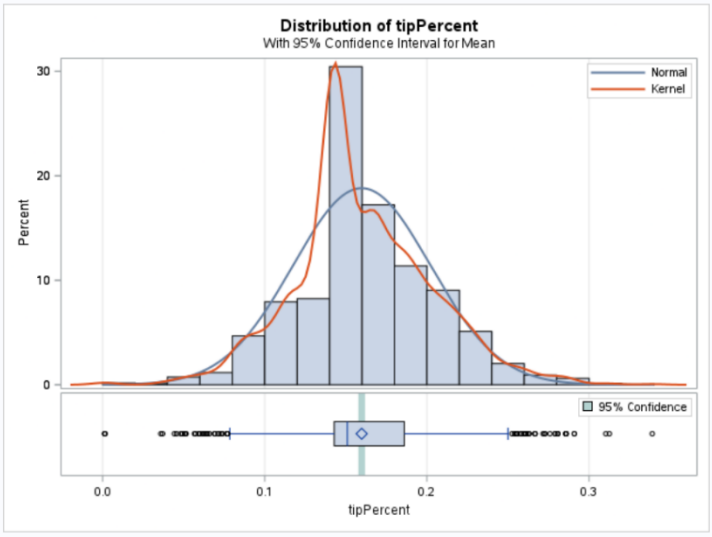
The TTEST Procedure

Variable: tipPercent

N	Mean	Std Dev	Std Err	Minimum	Maximum
1626	0.1598	0.0424	0.00105	0.00122	0.3390

Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0.1598	0.1577	0.1618	0.0424	0.0410	0.0439

DF	t Value	Pr > |t|
1625	9.30	<.0001



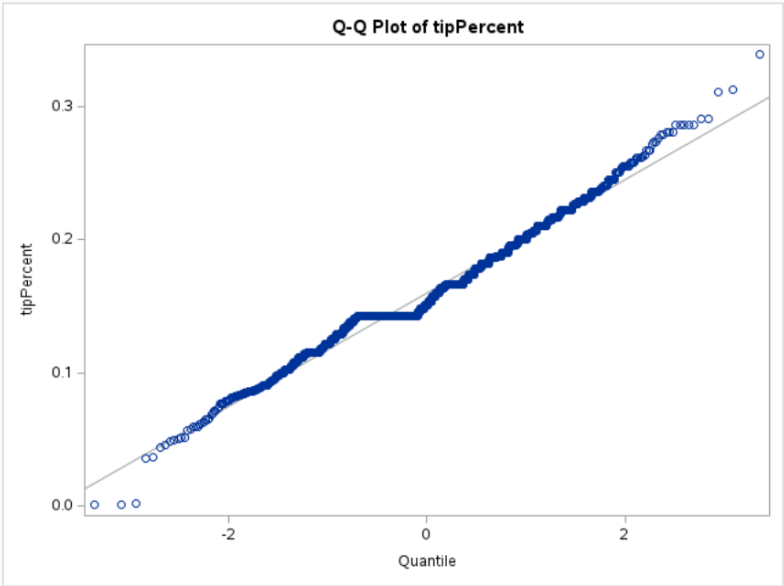


Figure 10

The TTEST Procedure

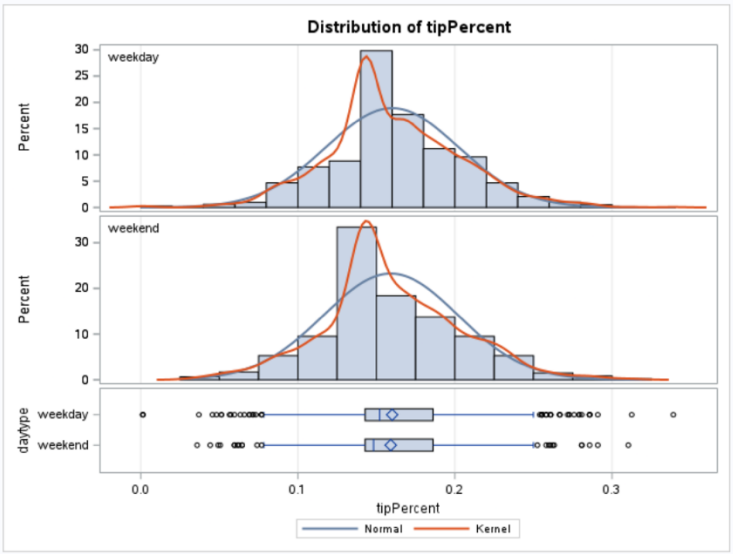
Variable: tipPercent

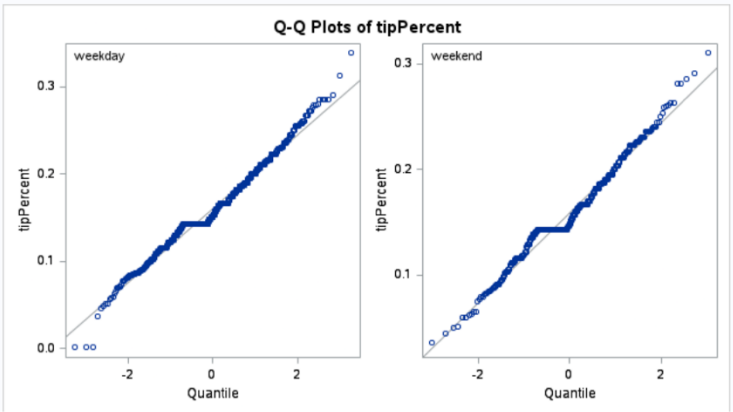
daytype	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
weekday		1153	0.1600	0.0422	0.00124	0.00122	0.3390
weekend		473	0.1591	0.0429	0.00197	0.0359	0.3103
Diff (1-2)	Pooled		0.000918	0.0424	0.00232		
Diff (1-2)	Satterthwaite		0.000918		0.00233		

daytype	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
weekday		0.1600	0.1576	0.1625	0.0422	0.0406	0.0440
weekend		0.1591	0.1552	0.1630	0.0429	0.0403	0.0458
Diff (1-2)	Pooled	0.000918	-0.00362	0.00546	0.0424	0.0410	0.0439
Diff (1-2)	Satterthwaite	0.000918	-0.00366	0.00550			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	1624	0.40	0.6918
Satterthwaite	Unequal	865.24	0.39	0.6938

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	472	1152	1.03	0.6618





II. Code

```
/* ++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++ */
/* STAT 403: Final Project Analyzing Chicago Taxi                           */
/* Coded by: Ammar Al-Zuhairi and Kellie Clinton                            */
/* Group Name: The Last Semicolon                                           */
/* ++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++ */
/* ------------------------------------------------------------------------ */
/* Loading The Data                                                         */
/* ------------------------------------------------------------------------ */
data taxi;
    infile "/home/u64426032/Project/Data/chicago_taxi.csv" dlm="," firstobs=2;
    input date DDMMYY10. daytype $ duration distance tip fare payment_type $ speed;
    format tip dollar8.2 fare dollar8.2 distance 5.2 speed 5.2 date DDMMYY10.;
run;

/* ------------------------------------------------------------------------ */
/* Modifing The Data                                                        */
/* ------------------------------------------------------------------------ */
data taxi;
    set taxi;
    where '01JAN2019'd <=date <='31MAR2019'd;
    totalAmount=fare + tip;
    format totalAmount dollar8.2;

    if tip > 0 then
        tipped="Yes";
    else
        tipped="No";
    tipPercent=tip / totalAmount;
run;

/* ------------------------------------------------------------------------ */
/* Fitting a Multiple Linear Regression.                                     */
/* ------------------------------------------------------------------------ */
/* Figure 1 & 2.                                                             */
/* ------------------------------------------------------------------------ */
proc reg data=taxi;
model fare = distance duration speed / vif;
run;
```

```
/* ---------------------------------------------------------------------- */
/* Checking variables distruption.                                         */
/* ---------------------------------------------------------------------- */
/* Figure 3.                                                               */
/* ---------------------------------------------------------------------- */
PROC UNIVARIATE DATA=taxi noprint;
   VAR fare distance duration speed;
   HISTOGRAM fare distance duration speed / NORMAL;
RUN;

/* ---------------------------------------------------------------------- */
/* Creating new log transformed variables.                                 */
/* ---------------------------------------------------------------------- */
data taxi;
set taxi;
lnFare = log(fare);
lnDistance = log(distance);
lnDuration = log(duration);
lnSpeed = log(speed);
run;

/* ---------------------------------------------------------------------- */
/* Fitting a Multiple Linear Regression With log transformed varibles      */
/* ---------------------------------------------------------------------- */
/* Figure 4.                                                               */
/* ---------------------------------------------------------------------- */
proc reg data=taxi;
model lnFare = lnDistance lnDuration lnSpeed  / vif;
run;

/* ---------------------------------------------------------------------- */
/* Selecting best Lambda for Ridge                                         */
/* ---------------------------------------------------------------------- */
/* Figure 5.                                                               */
/* ---------------------------------------------------------------------- */
proc reg data=taxi
    outvif
    plots=ridge(unpack)
    outest=taxi2
    ridge = 0 to 0.005 by 0.001;
model lnFare = lnDistance lnDuration lnSpeed / vif;
run;

proc print data=taxi2;
run;
```

```
/* ---------------------------------------------------------------------- */
/* Fitting Ridge Model                                                    */
/* ---------------------------------------------------------------------- */
/* Figure 6 & 7.                                                          */
/* ---------------------------------------------------------------------- */
proc reg data=taxi
    outest=taxi3;
model lnFare = lnDistance lnDuration lnSpeed /ridge = 0.002 outseb cli;
run;

proc print data=taxi3;
run;

/* ---------------------------------------------------------------------- */
/* Distribution of Tip Percentage Across Tipping customers                */
/* ---------------------------------------------------------------------- */
/* Figure 8.                                                              */
/* ---------------------------------------------------------------------- */
proc univariate data=taxi noprint;
    where tipped="Yes";
    var tipPercent;
    histogram tipPercent / vscale=count vaxislabel="Ride Count" midpoints=0 to .35
        by .05 normal;
    Inset mean="Average" std="St. Deviation" /position=NE;
    title "Distribution of Tip Percentage Across Tipping customers";
run;

/* ---------------------------------------------------------------------- */
/* Running a TTest for the tip percent                                    */
/* ---------------------------------------------------------------------- */
/* Figure 9.                                                              */
/* ---------------------------------------------------------------------- */
proc ttest data=taxi h0=0.15;
where tipped = "Yes";
var tipPercent;
Run;

/* ---------------------------------------------------------------------- */
/* Running a TTest for the tip percent With day type class                */
/* ---------------------------------------------------------------------- */
/* Figure 10.                                                              */
/* ---------------------------------------------------------------------- */
proc ttest data=taxi;
where tipped = "Yes";
class daytype;
var tipPercent;
Run;
```